

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

This project applied a full end-to-end data science workflow to understand and predict the reusability of SpaceX's Falcon 9 first stage—an essential driver of launch cost and competitive advantage in the commercial space industry. By combining data engineering, exploratory analytics, and predictive modeling, the work produced both strategic insights and an operational prediction tool that a real aerospace company could use.

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Executive Summary – Summary of Methodologies

- **Data Acquisition & Integration:** Collected launch records from multiple public sources, merged them into a unified dataset, and validated completeness and consistency.
- **Data Wrangling & Feature Engineering:** Cleaned missing values, standardized categorical variables, engineered features such as booster version categories, and prepared the dataset for modeling.
- **Exploratory Data Analysis (EDA):** Used SQL queries, statistical summaries, and visual dashboards to uncover patterns across launch sites, payload ranges, booster generations, and landing outcomes.
- **Interactive Visualization & Dashboards:** Built dynamic charts (pie charts, scatter plots, filters, sliders) to help stakeholders explore launch success distribution, payload relationships, and site performance.
- **Predictive Modeling & Evaluation:** Trained multiple classification models (Logistic Regression, SVM, KNN, Decision Tree), tuned hyperparameters, compared accuracy scores, and validated performance using confusion matrices.

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Executive Summary – Results

- Launch Site Insights: KSC LC-39A demonstrated the strongest success ratio, confirming its role as SpaceX's most reliable landing site.
- Payload & Booster Findings: Newer booster versions (especially B5) delivered the highest success rates, even at heavier payloads—showing clear technological progression.
- Model Performance: The Decision Tree classifier achieved the highest accuracy (88.89%), outperforming all other models and demonstrating strong predictive capability for landing outcomes.
- Operational Value: The final model provides a practical mechanism for estimating whether a first stage will land, enabling cost estimation and strategic planning for a hypothetical competitor like "SpaceY."
- End-to-End Deliverable: The project produced a complete analytical pipeline—from raw data to dashboards to a deployable predictive model—mirroring real industry workflows.

Introduction

- Project background and context

Commercial spaceflight is shifting toward reusable rockets, with SpaceX leading the way by recovering the Falcon 9 first stage and dramatically lowering launch costs. As Space Y enters this market, understanding when a booster is likely to land becomes essential for pricing and competitive strategy. This project uses public SpaceX data and machine learning to predict first-stage landing outcomes.

- Problems you want to find answers

1. Can we predict whether the Falcon 9 first stage will land successfully?
2. Which mission factors influence landing outcomes the most?
3. How can these predictions help Space Y estimate launch costs and compete effectively?

Section 1

Methodology

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Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

Data sets were collected by

1. API Data
2. Web Scraping (Supplementary Source)
3. Data Cleaning & Preparation

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Data Collection – SpaceX API

SpaceX REST API (Primary Source)

- Queried the SpaceX v4 API for past launch records
- Used GET requests to retrieve structured JSON
- Flattened nested JSON with `json_normalize`
- Enriched each launch by calling secondary endpoints (Booster, Launchpad, Payload, Core) using ID fields

[SpaceX API Notebook](#)

Start



GET /launches/past

(Retrieve past launch JSON)



Normalize JSON

(Flatten with `json_normalize`)



Extract IDs

(booster, payload, launchpad, core)



Call Secondary Endpoints

`/rockets` • `/payloads` • `/launchpads` • `/cores`



Merge All Data



Filter Falcon 9 Only



Handle Missing Values



Final Clean Dataset

Data Collection - Scraping

- Scraped Falcon 9 launch tables from relevant Wikipedia pages
- Parsed HTML structure using BeautifulSoup
- Converted extracted tables into Pandas Data
- FramesUsed scraped records to validate and supplement API data

[Web scraping Notebook](#)

Start



Load Wikipedia Page



Parse HTML with BeautifulSoup



Locate Launch Tables



Extract Rows & Columns



Convert to Pandas DataFrame



Cross-check with API Data



Enhanced Dataset

Data Wrangling

- Selected key launch attributes (payload, orbit, booster, site, outcome)
- Cleaned and standardized categories
- Converted landing outcomes into a binary target (0/1)
- Filtered out non-Falcon 9 missions
- Handled missing values (mean-filled payload mass; kept meaningful nulls)
- Combined and refined fields into a clean, modeling-ready dataset

[Data Wrangling Notebook](#)

Start



Load Raw Data

(API + Web-scraped tables)



Select Key Attributes



Clean & Standardize Categories



Convert Outcome → Binary (0/1)



Filter Falcon 9 Only



Handle Missing Values

(PayloadMass → mean; keep meaningful nulls)

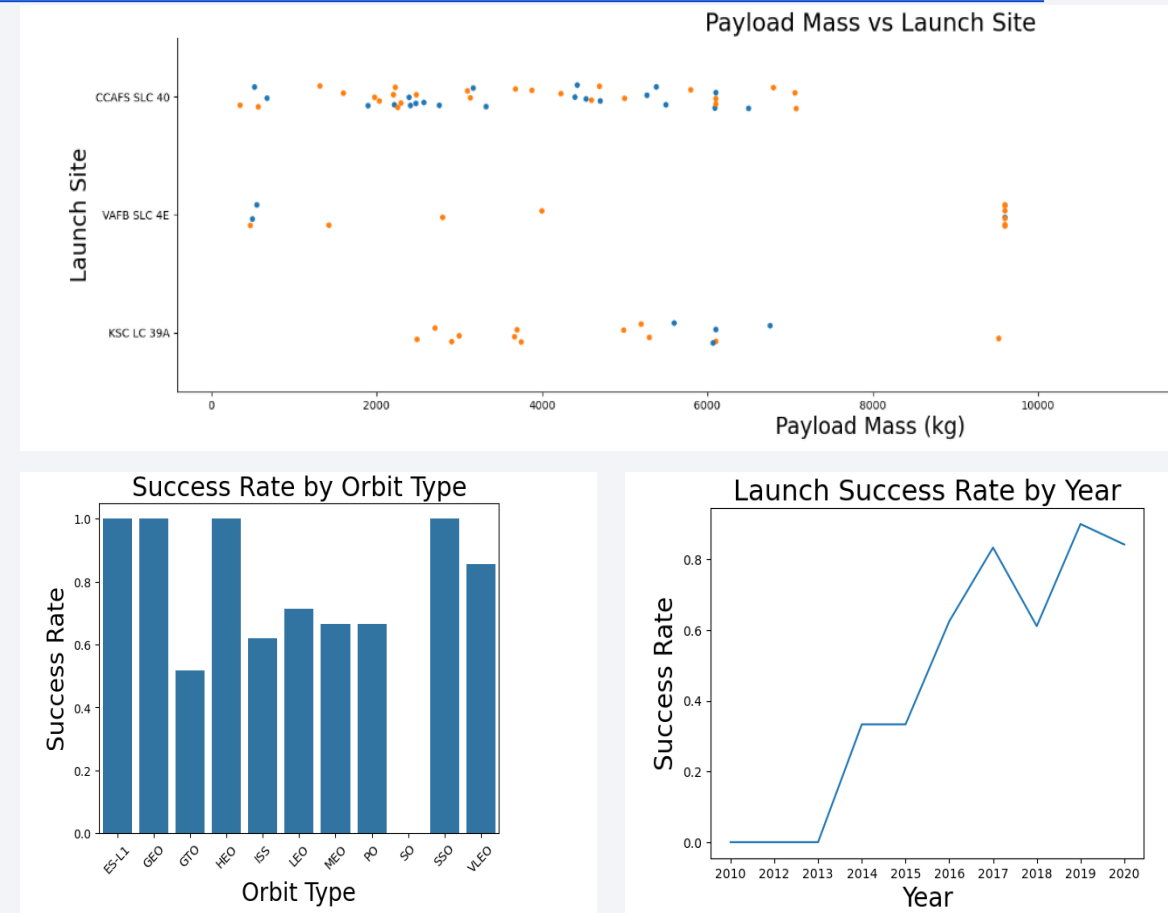


Finalize Clean Dataset

EDA with Data Visualization

- Catplots & scatter plots helped explore relationships between categorical variables (site, orbit) and numerical ones (flight number, payload).
- Bar charts made it easy to compare average success rates across categories.
- Line chart showed improvement over time — a key insight for SpaceX's reliability.

[EDA Visualization Notebook](#)



EDA with SQL

Summary of Queries Performed

- Retrieved all unique launch sites to understand the full set of locations used in SpaceX missions.
- Calculated the total payload mass for NASA (CRS) missions, focusing on boosters supporting resupply operations.
- Computed the average payload mass for booster version “F9 v1.1” using an aggregate function.
- Identified the earliest successful ground-pad landing by selecting the minimum date with that outcome.
- Listed boosters with successful drone-ship landings carrying payloads between 4000 and 6000 kg.
- Counted total successful and failed mission outcomes to understand the overall distribution of landing results.
- Found booster versions that carried the maximum payload mass using a subquery to identify the highest payload value.
- Extracted 2015 records with drone-ship failures, including month, landing outcome, booster version, and launch site.
- Ranked landing outcomes between 2010-06-04 and 2017-03-20 in descending order based on their frequency. Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

[EDA with SQL Notebook](#)

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Build an Interactive Map with Folium

Folium Map Objects

- **Markers** - Added markers at each launch site to pinpoint the exact geographic locations of SpaceX pads.
- **Circle Markers** - Placed circles around launch sites to highlight the surrounding area and make each site visually stand out.
- **Popups / Tooltips** - Attached popups to markers so clicking reveals site names and basic launch information without cluttering the map.
- **Lines (Polylines)** - Drew lines between launch and landing points to show spatial relationships and booster travel paths.

Why These Objects Were Added

- Markers make each launch site easy to identify.
- Circles help emphasize the region around each site.
- Popups provide quick context without overwhelming the map.
- Lines visually connect launch and landing locations, helping explain mission geography.

[Interactive Map and Folium Notebook](#)

Build a Dashboard with Plotly Dash

Dashboard — Plots and Interactions

- **Launch Site Dropdown** - Lets users switch between *all sites* or a *specific site* to instantly update the visuals.
- **Pie Chart (Success Distribution)** - Shows overall success rate or site-specific success rate depending on the dropdown selection.
- **Payload Range Slider** - Allows users to filter launches by payload mass and explore how payload affects landing success.
- **Scatter Plot (Payload vs. Landing Outcome)** - Displays the relationship between payload mass and landing success, colored by booster version, and updates based on both dropdown and slider inputs.

Why These Were Added

- To make the dashboard **interactive**, not static.
- To let users **compare launch sites**, **filter by payload**, and **see patterns** in landing success.
- To help users explore **how payload, site, and booster version** influence mission outcomes in a simple, visual way.

[Plotly Dash Notebook](#)

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Predictive Analysis (Classification)

Predictive Analysis — Summary of the Model Development Process

- How We Built the Models
- Prepared the dataset by selecting relevant features and encoding categorical variables.
- Split the data into training and test sets to ensure fair evaluation.
- Trained multiple classification models, including:
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

[ML Predictive Analytics Notebook](#)

Flowchart — Model Development Process

Start



Data Preparation

(feature selection, encoding, train–test split)



Train Baseline Models

(Logistic Regression, SVM, Decision Tree, KNN)



Evaluate Baseline Performance

(accuracy + confusion matrix)



Hyperparameter Tuning

(GridSearchCV for each model)



Retrain With Best Parameters



Evaluate Tuned Models



Compare All Models



Select Best Classifier

(highest accuracy + best error balance)

Predictive Analysis (Classification)

How We Evaluated the Models

- Measured accuracy on the test set for each model.
- Generated confusion matrices to understand true/false positives and negatives.
- Compared performance across all models to identify strengths and weaknesses.

[ML Predictive Analytics Notebook](#)

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Predictive Analysis (Classification)

How We Improved the Models

- Applied GridSearchCV to tune hyperparameters for each classifier.
- Optimized parameters such as:
- C and kernel type (SVM)
- Depth and splitting criteria (Decision Tree)
- Number of neighbors (KNN)
- Regularization strength (Logistic Regression)
- Retrained models using the best hyperparameters.

[ML Predictive Analytics Notebook](#)

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Predictive Analysis (Classification)

How We Found the Best Model

- How We Found the Best Model
- Evaluated tuned models again using accuracy and confusion matrices.
- Compared all optimized models side-by-side.
- Selected the best-performing classifier based on highest accuracy and most balanced error distribution.

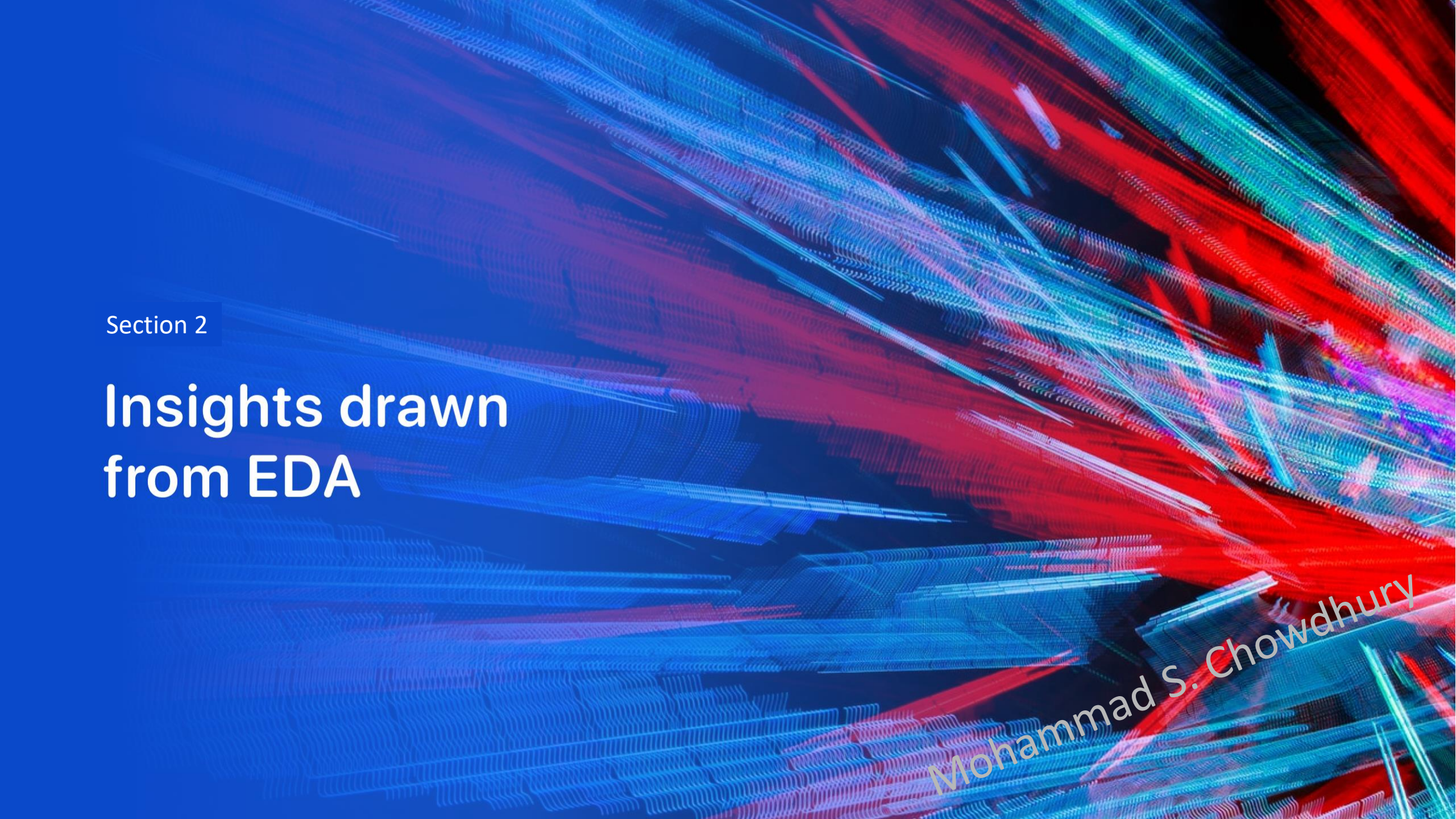
[ML Predictive Analytics Notebook](#)

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Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

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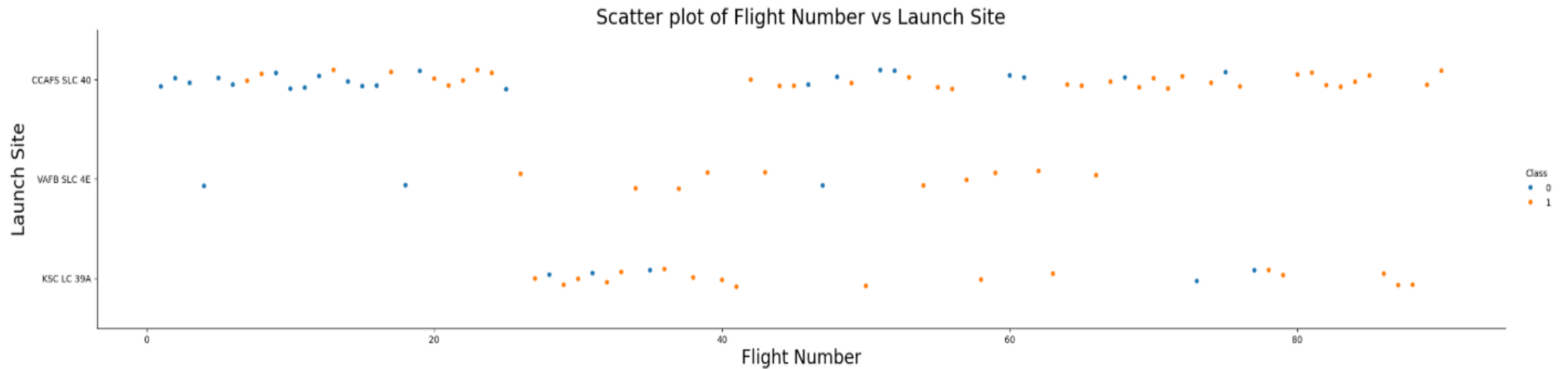


Section 2

Insights drawn from EDA

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Flight Number vs. Launch Site



Flight Number vs. Launch Site — Pattern Explanation

Early flights show more failures

In all launch sites, the lower flight numbers are mostly blue points (Class 0). This reflects SpaceX's early learning curve — the first missions had a higher chance of failure.

Success increases with experience

As flight numbers rise, orange points (Class 1) become more common. This shows that landing reliability improves as SpaceX gains more operational experience.

Launch sites behave differently

Each site has its own mix of successes and failures.

- **CCAFS SLC-40** shows a more scattered mix of outcomes.
- **KSC LC-39A** and **VAFB SLC-4E** show more orange points overall.
This means launch site contributes to landing success.

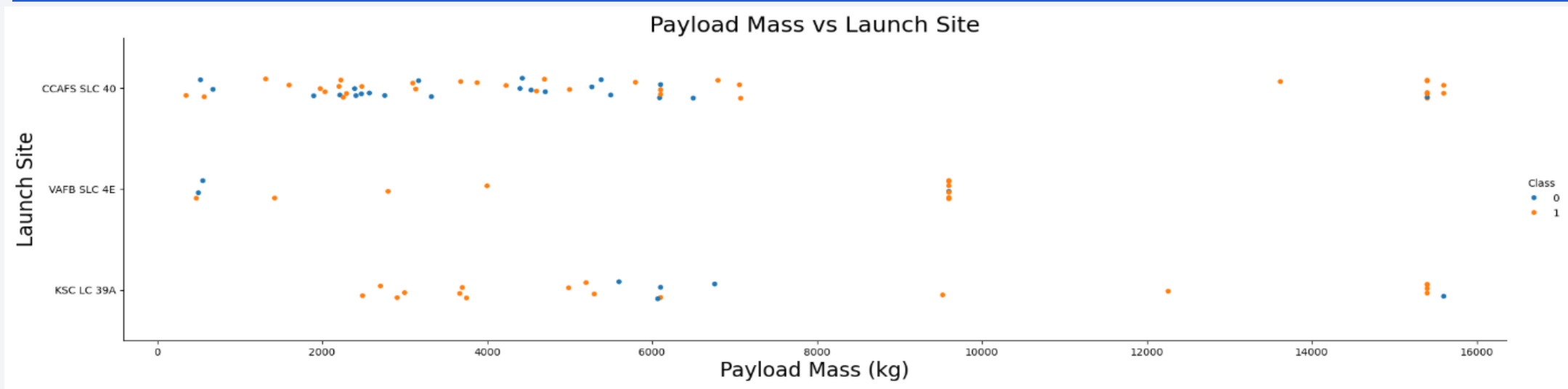
Clear visual clusters

Higher-numbered flights at KSC LC-39A and VAFB SLC-4E form clusters of orange points, while CCAFS SLC-40 has more mixed clusters.

These clusters help reveal which sites consistently perform better.

The scatter plot shows that **launch site and flight experience both influence landing success**. Early missions fail more often, and some sites consistently achieve better outcomes than others.

Payload vs. Launch Site



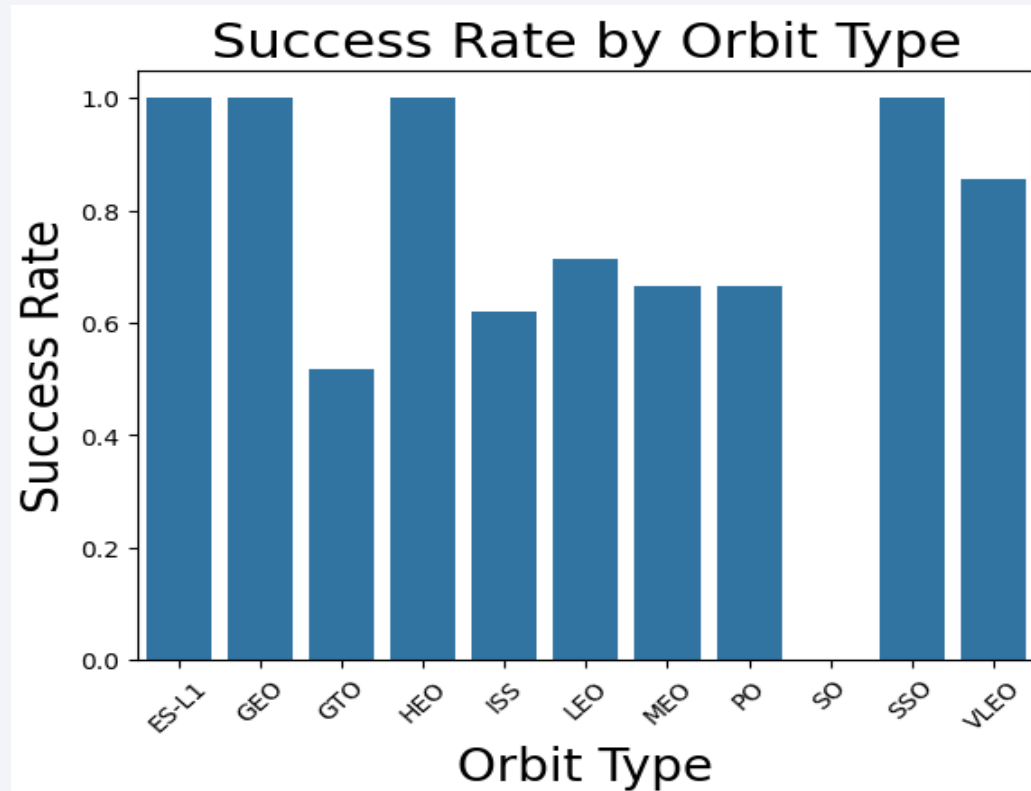
Payload Mass vs. Launch Site — Explanation

When you look at this scatter plot, a few things jump out right away:

- Each launch site handles all kinds of payload sizes, but **KSC LC-39A** clearly takes on many of the heavier missions — and it handles them well.
- **CCAFS SLC-40** is more unpredictable. You see a mix of successes and failures across the whole payload range.
- What's interesting is that even the **really heavy payloads** — the ones above 10,000 kg — still land successfully. That tells us that "heavy" doesn't automatically mean "risky."
- You can also see little clusters of orange (successful landings) forming at certain sites, especially KSC LC-39A and VAFB SLC-4E. Those clusters make it obvious that some locations consistently perform better.

Overall, the plot makes one thing clear: **launch site and payload mass together shape the landing outcome**, and you can literally see that pattern forming in the dots.

Success Rate vs. Orbit Type



Success Rate by Orbit Type — Explanation

This chart shows how reliable SpaceX missions are depending on the orbit they target:

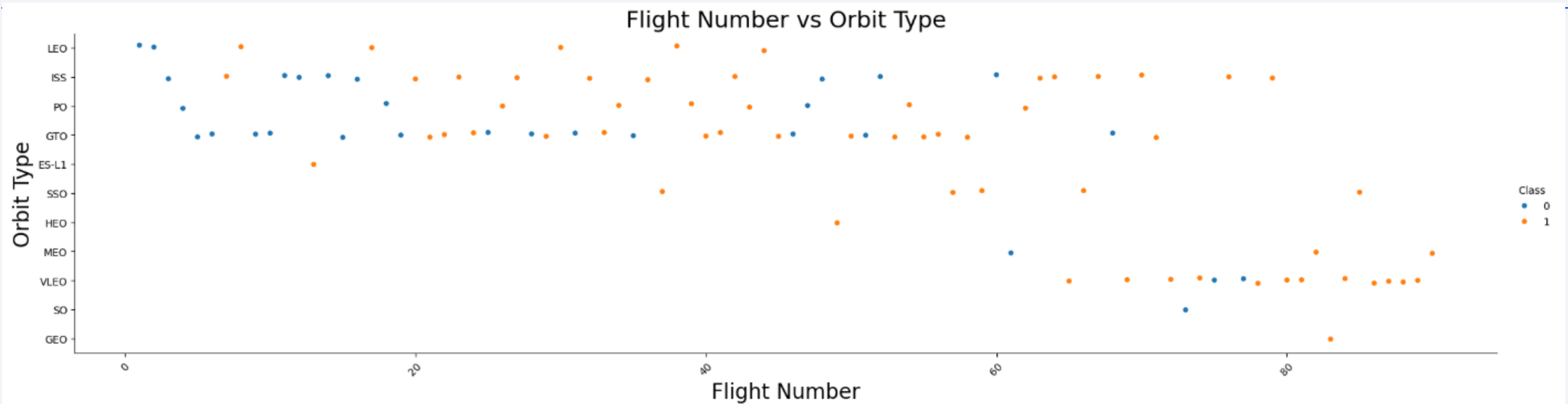
- **Some orbits are nearly perfect:** ES-L1, GEO, HEO, and SSO all have success rates close to 100%. These missions are rare but consistently successful.
- **GTO stands out as risky:** It has the **lowest success rate**, around 55%, making it the most failure-prone orbit in this dataset.
- **Mid-range orbits vary:** ISS, LEO, MEO, PO, and SO sit between 62% and 70% — not bad, but not as reliable as the top performers.
- **VLEO performs well:** With a success rate above 85%, it's one of the more dependable orbit types.

Orbit type clearly affects landing success.

Some orbits are consistently safe, while others — like GTO — carry more risk.

This insight helps us choose better features when building predictive models.

Flight Number vs. Orbit Type



Flight Number vs. Orbit Type — Humanized Explanation

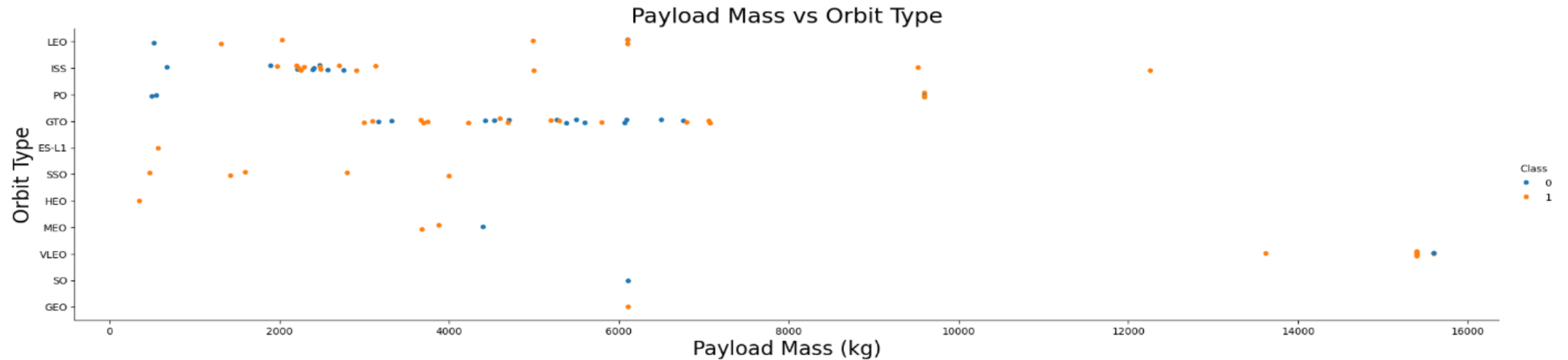
This scatter plot shows how SpaceX missions evolved over time across different orbit types:

- **Early flights were riskier** - Lower flight numbers show more blue dots (Class 0), meaning more failed landings in the beginning.
- **Success improves with experience** - As flight numbers increase, orange dots (Class 1) become more common — showing that SpaceX got better at landing over time.
- **Orbit type influences outcome** - Some orbits like **SSO, GEO, and ES-L1** show mostly successful landings, while **GTO** has more failures — matching what we saw in the orbit success rate chart.

Bottom Line

The plot shows that **flight experience and orbit type both affect landing success**. Combining these features helps build a smarter prediction model.

Payload vs. Orbit Type



Payload Mass vs. Orbit Type — Humanized Explanation

This plot shows how different orbit types handle different payload sizes — and how successful those missions are:

- **Payloads range widely across orbits**

Some orbits like **LEO, ISS, and GTO** carry everything from light to heavy payloads. Others like **SSO, GEO, and ES-L1** tend to carry more specific ranges.

- **Success isn't just about weight**

You'll see **orange dots (Class 1)** even at the highest payloads — meaning heavy missions can still land successfully.

- **Orbit type matters**

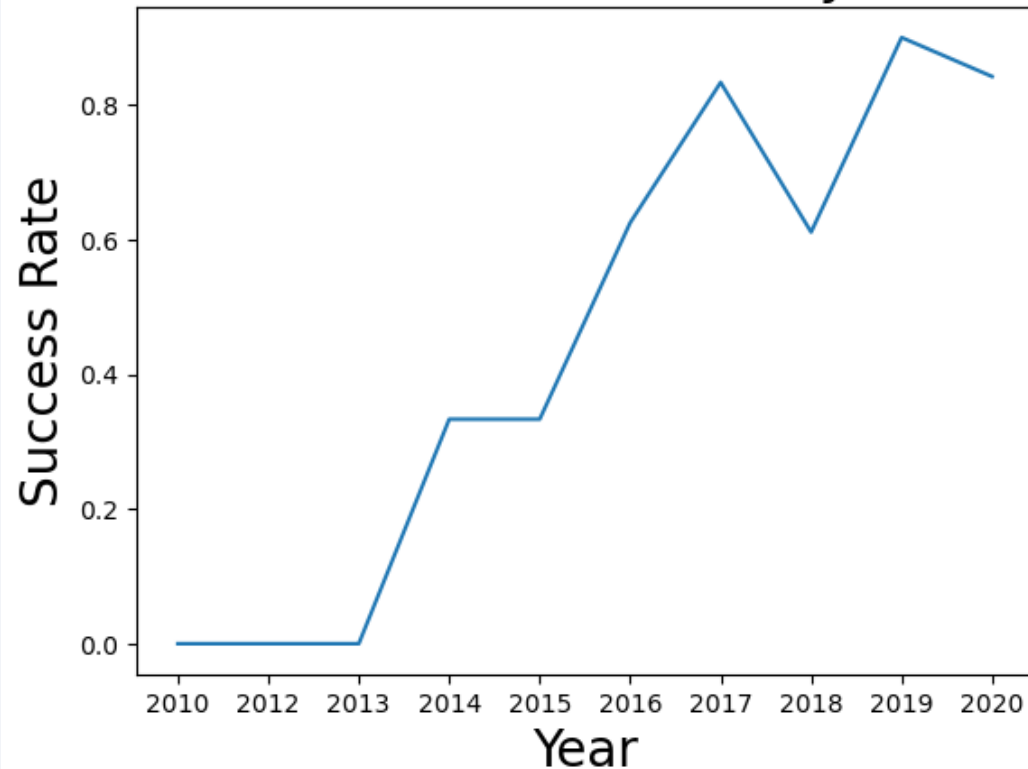
Certain orbits like **SSO, GEO, and ES-L1** show mostly successful landings, while **GTO** has more failures — matching the orbit success rate chart.

Bottom Line

The scatter plot shows that **orbit type and payload mass together influence landing success**. Some orbits handle heavy payloads better, and success isn't just about size — it's about the mission profile.

Launch Success Yearly Trend

Launch Success Rate by Year



Launch Success Rate by Year — Explanation

This graph tells the story of SpaceX's journey over time:

- **Early years were rough**

From 2010 to 2013, the success rate was zero — no successful landings yet.

- **Steady improvement begins**

In 2014 and 2015, success climbs to around 33%, then jumps to 63% in 2016.

- **Breakthrough in 2017**

Success peaks at 85%, showing major progress in reliability.

- **A dip, then recovery**

2018 sees a drop to 62%, but by 2019 it rebounds to 90%.

2020 dips slightly again, but overall performance remains strong.

Bottom Line

The line graph shows how **SpaceX learned from early failures**, improved rapidly, and became a consistently reliable launch provider.

The growth is made by setbacks, better technology and engineering, more data, smarter decisions and long-term success.

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All Launch Site Names

- Find the names of the unique launch sites

```
%%sqlSELECT DISTINCT "Launch_Site"FROM SPACEXTBL;
```

Query results

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

This query extracts all unique launch site names from the dataset. It helps us understand where SpaceX missions are launched from, which is important for analyzing site-specific success rates and patterns.

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

```
%%sqlSELECT *FROM SPACEXTBLWHERE "Launch_Site" LIKE  
'CCA%'LIMIT 5;
```

Query results

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Filtering by launch site helps us analyze site-specific performance, identify patterns in mission outcomes, and understand how location influences success rates. This is a key step in building predictive models.

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

```
%%sql
SELECT SUM("PAYLOAD_MASS_KG_") AS Total_Payload_Mass
FROM SPACEXTBL
WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Payload_Mass

45596

This query adds up the payload mass from all launches where the customer was NASA (CRS). It shows that SpaceX boosters have carried a total of 45,596 kilograms for NASA's CRS missions — a strong indicator of the scale and reliability of these cargo deliveries to the ISS.

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%%sql
```

```
SELECT AVG("PAYLOAD_MASS__KG_") AS Average_Payload_Mass  
FROM SPACEXTBL  
WHERE "Booster_Version" = 'F9 v1.1';
```

Query result: 2928.4

The result — 2928.4 kg — gives us a sense of the typical mission size for this booster type, useful for comparing performance across different versions.

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%%sql
```

```
SELECT MIN("Date") AS First_Ground_Pad_Success  
FROM SPACEXTBL  
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

Query result, First_Ground_Pad_Success: 2015-12-22

This query finds the earliest date when SpaceX successfully landed a booster on a ground pad. The result — December 22, 2015 — marks a major milestone in reusable rocket technology and SpaceX's landing achievements.

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%%sql
```

```
SELECT "Booster_Version"  
FROM SPACEXTBL  
WHERE "Landing_Outcome" = 'Success (drone  
      AND "PAYLOAD_MASS__KG_" > 4000  
      AND "PAYLOAD_MASS__KG_" < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

It highlights specific F9 Full Thrust versions that handled medium-heavy missions and still achieved precise landings — a sign of SpaceX's growing reliability in complex recoveries.

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

```
%%sql
```

```
SELECT TRIM("Mission_Outcome") AS Mission_Outcome, COUNT(*) AS  
Total_Count
```

```
FROM SPACEXTBL
```

```
GROUP BY TRIM("Mission_Outcome");      # To clean dirty data
```

Query result

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

This query counts how many missions ended in success or failure.

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

- `%%sql`

- `SELECT "Booster_Version", "PAYLOAD_MASS_KG_"`

- `FROM SPACEXTBL`

- `WHERE "PAYLOAD_MASS_KG_" = (`

- `SELECT MAX("PAYLOAD_MASS_KG_")`

- `FROM SPACEXTBL`

- `);`

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Each of these Falcon 9 Block 5 (F9 B5) boosters successfully launched missions with 15,600 kg payloads, showcasing their heavy-lift capability and engineering consistency.

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Query

```
%%sql
SELECT
    substr("Date", 6, 2) AS Month,
    "Booster_Version",
    "Launch_Site",
    "Landing_Outcome"
FROM SPACEXTBL
WHERE substr("Date", 1, 4) = '2015'
    AND "Landing_Outcome" = 'Failure (drone ship)';
```

Query results

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

It reveals that two boosters — B1012 and B1015 — launched from CCAFS LC-40 in January and April, but both failed to land on the drone ship. This helps pinpoint early recovery challenges before SpaceX's landing technology matured.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Query

```
%%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS Outcome_Count
FROM SPACEXTBL
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Outcome_Count DESC;
```

Query results

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

It shows that most missions during this period either had no landing attempt or landed on a drone ship — with both success and failure outcomes.

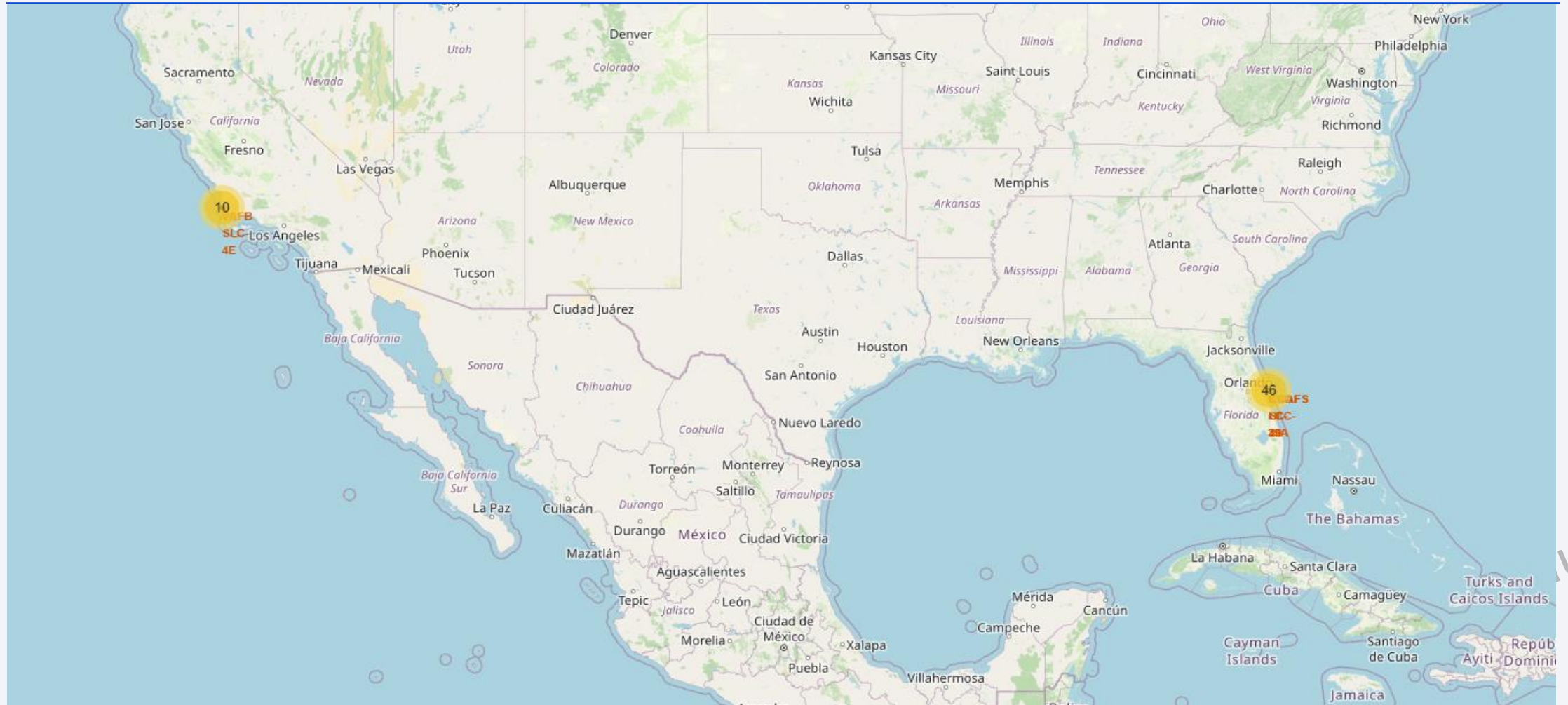


Section 3

Launch Sites Proximities Analysis

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Discovering SpaceX Using Folium



Discovering SpaceX Using Folium

Launch Site Markers:

Each marker represents a SpaceX launch site, such as:

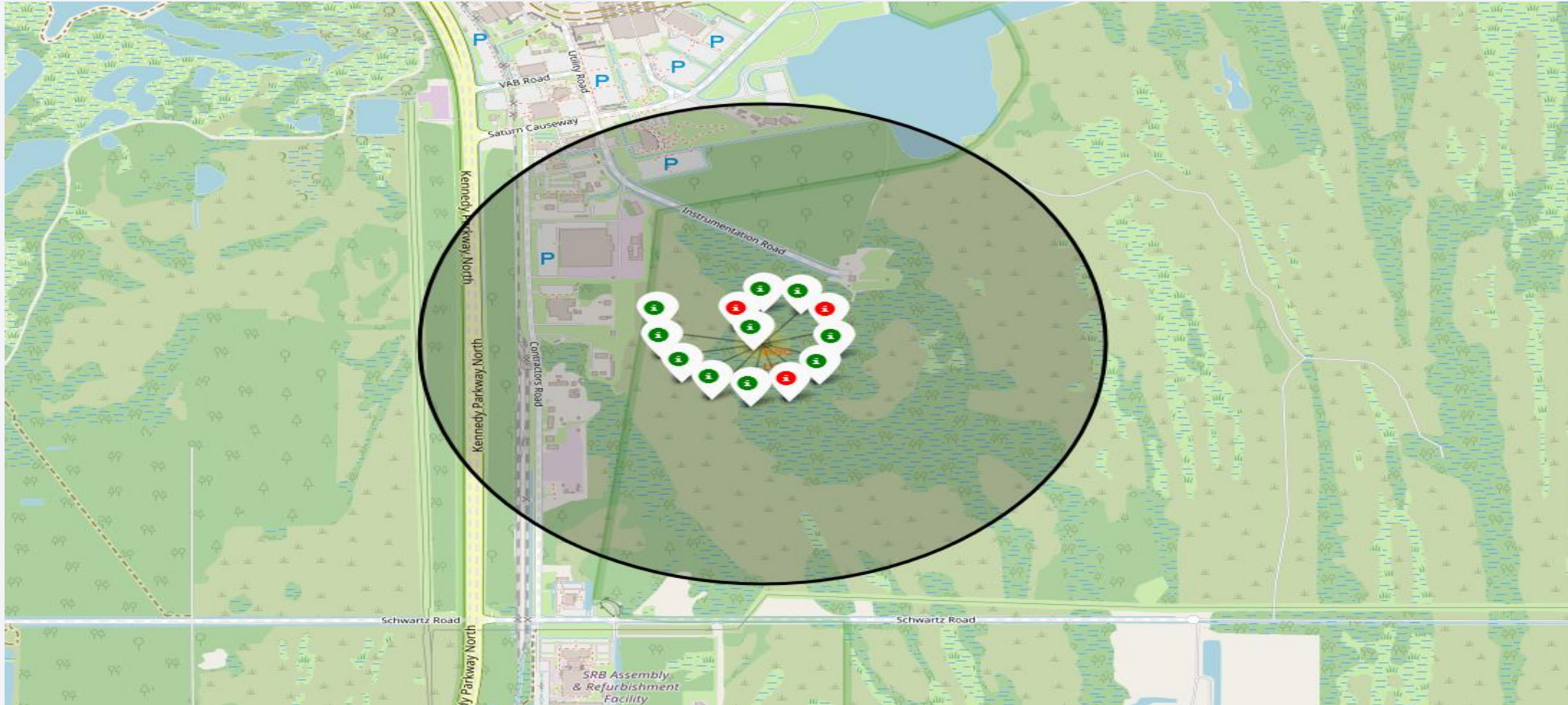
- **CCAFS LC-40** and **KSC LC-39A** in Florida
- **VAFB SLC-4E** in California

Geographic Spread: Most launch sites are concentrated in the United States, showing SpaceX's operational base.

Strategic Locations: Sites are near coastlines — ideal for eastward launches and safe booster recovery zones.

Global Context: The map helps visualize how launch geography affects orbital trajectories, recovery logistics, and mission planning.

Launch Outcome Visualization by Location



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Launch Outcome Visualization by Location

Green markers represent successful landings.

Red markers indicate failed landings.

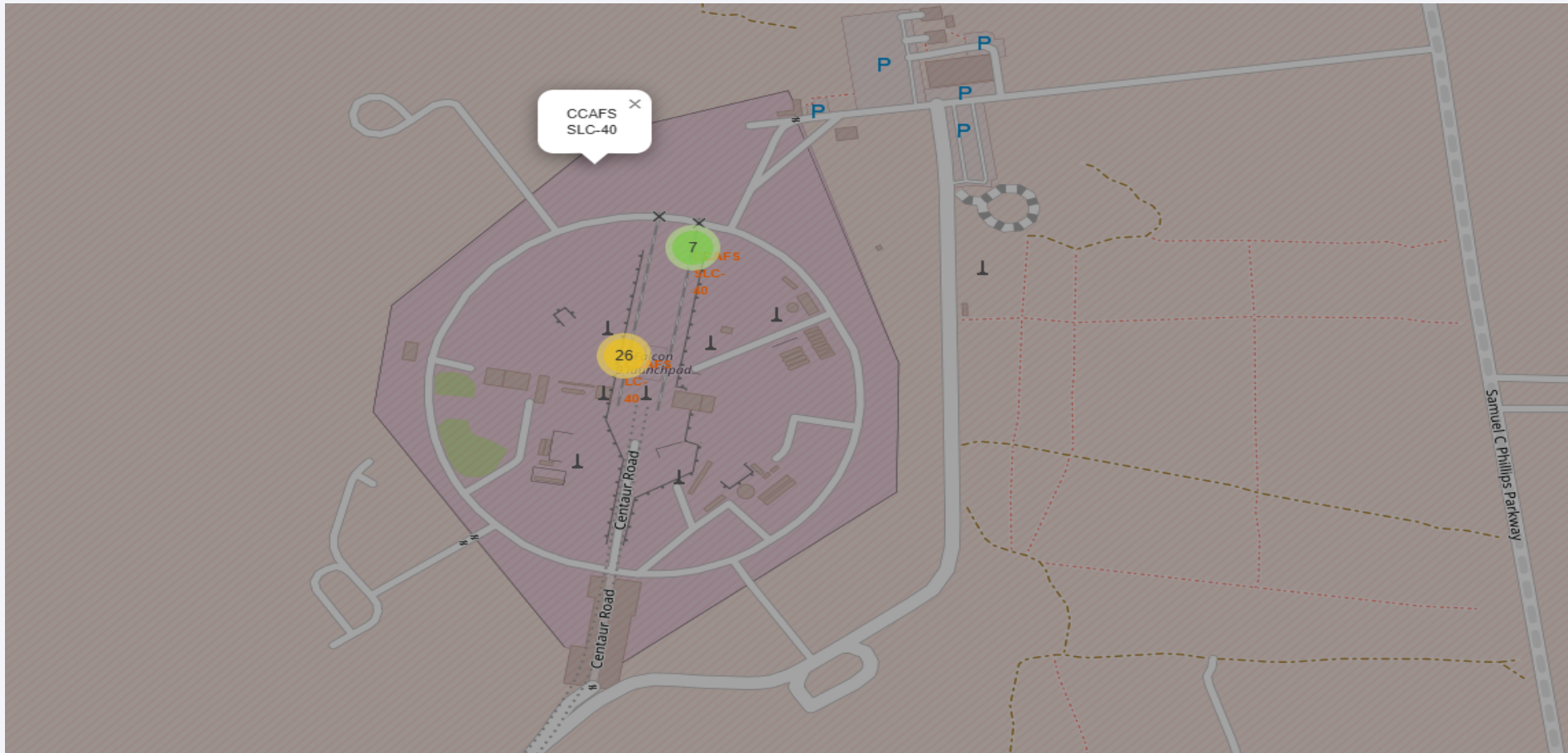
Clustered layout:

- The heart-shaped arrangement suggests a dense concentration of launches in a specific zone — likely near Kennedy Space Center.

Geographic context:

- Roads like **Saturn Causeway** and facilities like **SRB Assembly** confirm this is a high-activity launch and recovery area.

Spatial Analysis of CCAFS SLC-40 Launch Site



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Spatial Analysis of CCAFS SLC-40 Launch Site

CCAFS SLC-40 is clearly marked, often with a yellow or green icon.

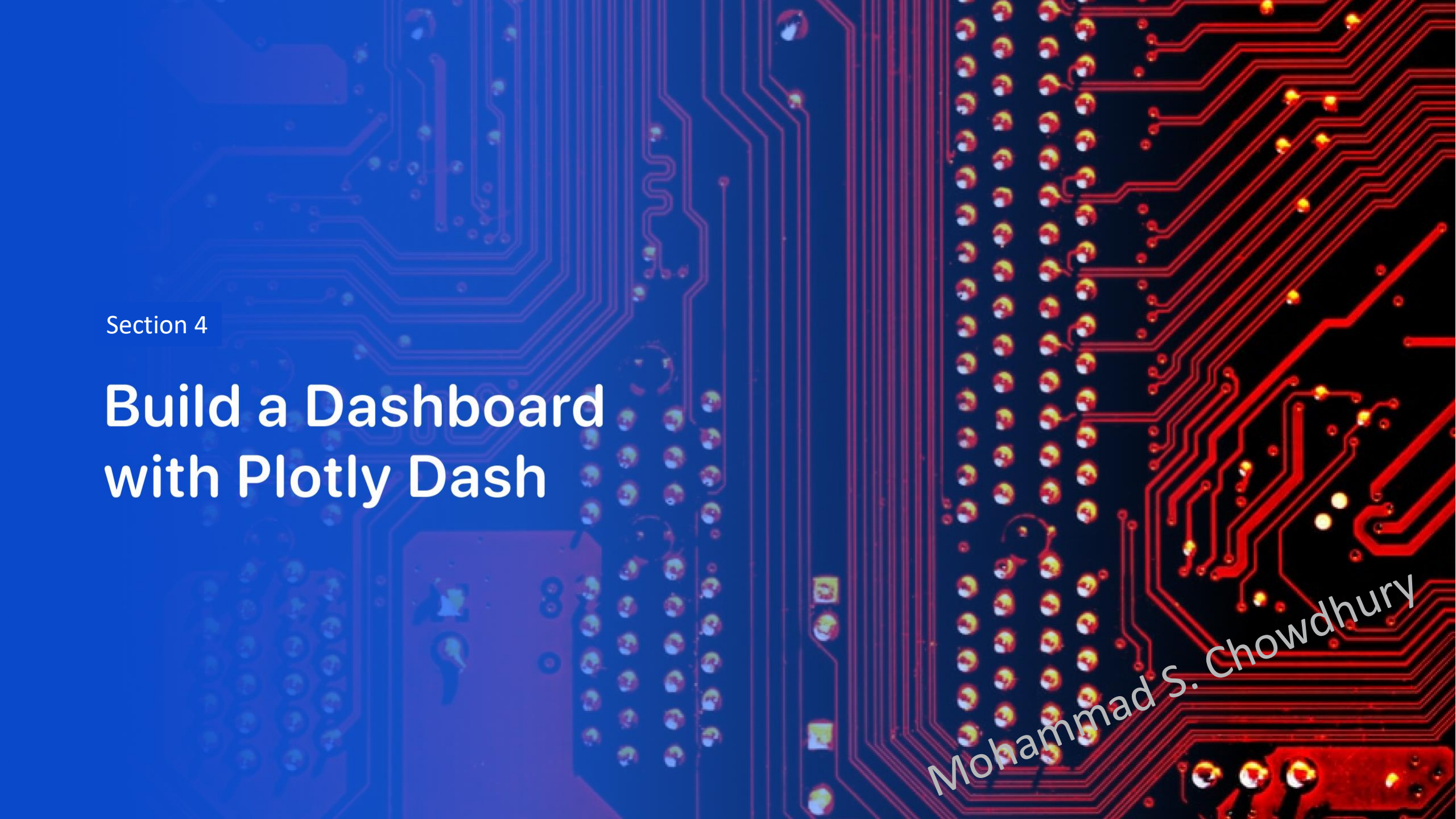
Proximity to Coastline: The site is **less than 3 km from the Atlantic Ocean**, ideal for eastward launches and safe booster recovery.

Highway Access: **Samuel C Phillips Parkway** and **Centaur Road** provide direct logistical access for equipment and personnel.

Railway (if mapped): If visible, rail lines support heavy cargo transport — crucial for booster and payload delivery.

Strategic Location: The site's placement balances **launch safety**, **transport efficiency**, and **environmental clearance**.

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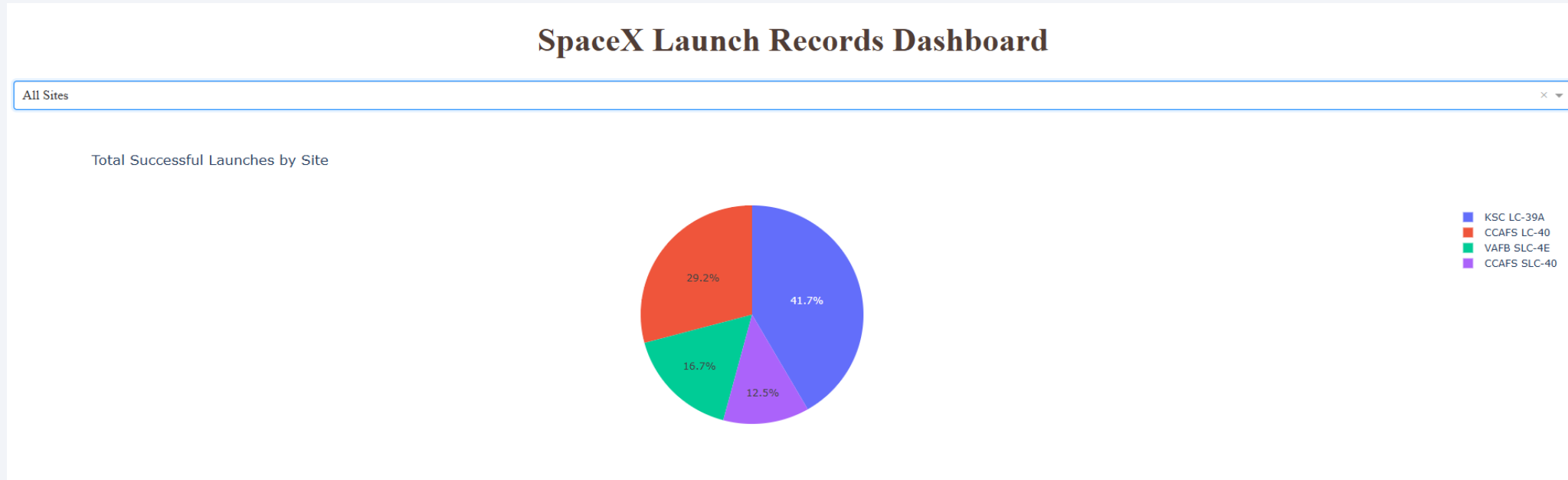


Section 4

Build a Dashboard with Plotly Dash

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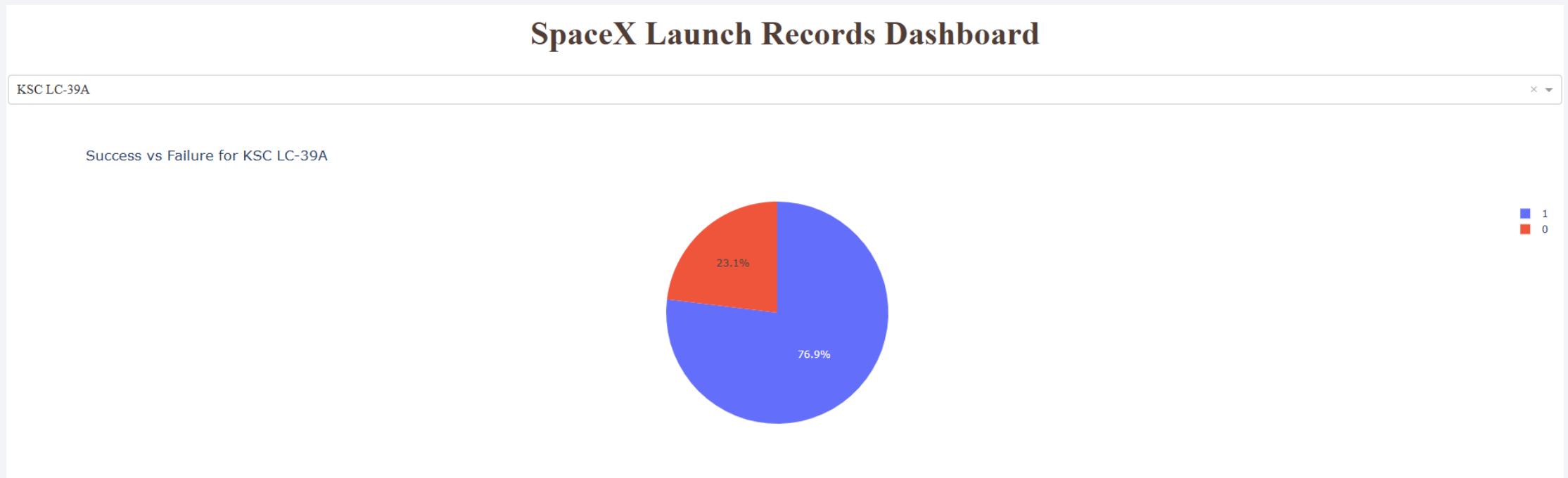
Success of All SpaceX Launches by Site



Dropdown Menu ("All Sites") - Allows filtering the data by individual launch site or viewing all sites together.

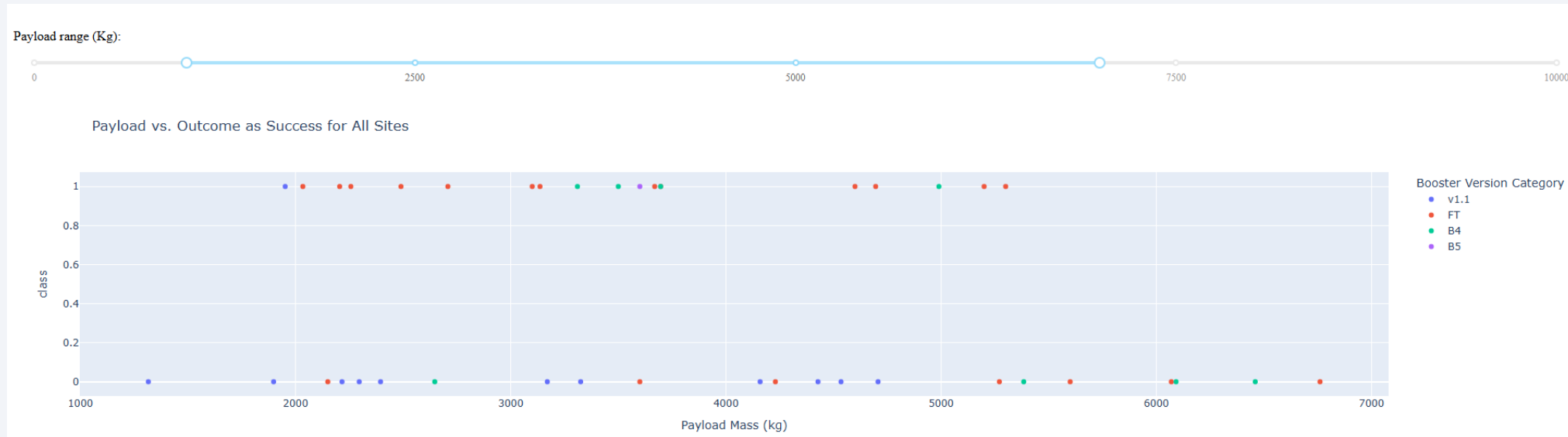
Pie Chart: "Total Successful Launches by Site" Visually breaks down the proportion of successful launches from each site.

Launch Success vs Failure Ratio — KSC LC-39A



- Blue Segment (76.9%) — Represents successful launches (coded as “1”)
- Red Segment (23.1%) — Represents failed launches (coded as “0”)
- Legend and Labels: The chart uses clear color coding and percentage labels to distinguish outcomes.

Payload Mass vs. Launch Success of All Sites



- F9 B5 (purple) shows the highest success rate, especially at high payloads (6000–10000 kg).
- Mid Payloads (3000–6000 kg): FT and B5 dominate with consistent success..

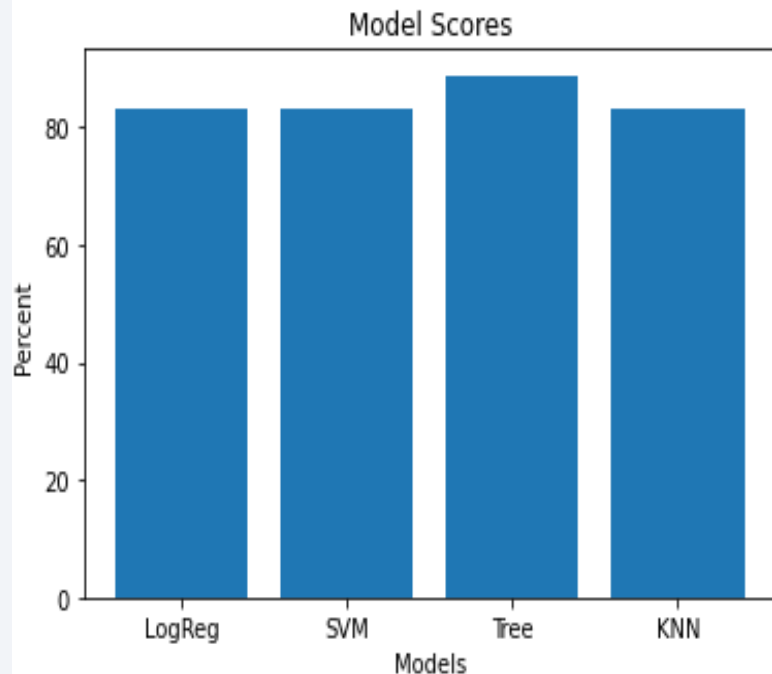
Section 5

Predictive Analysis (Classification)

Mohammad S. Chowdhury

Classification Accuracy

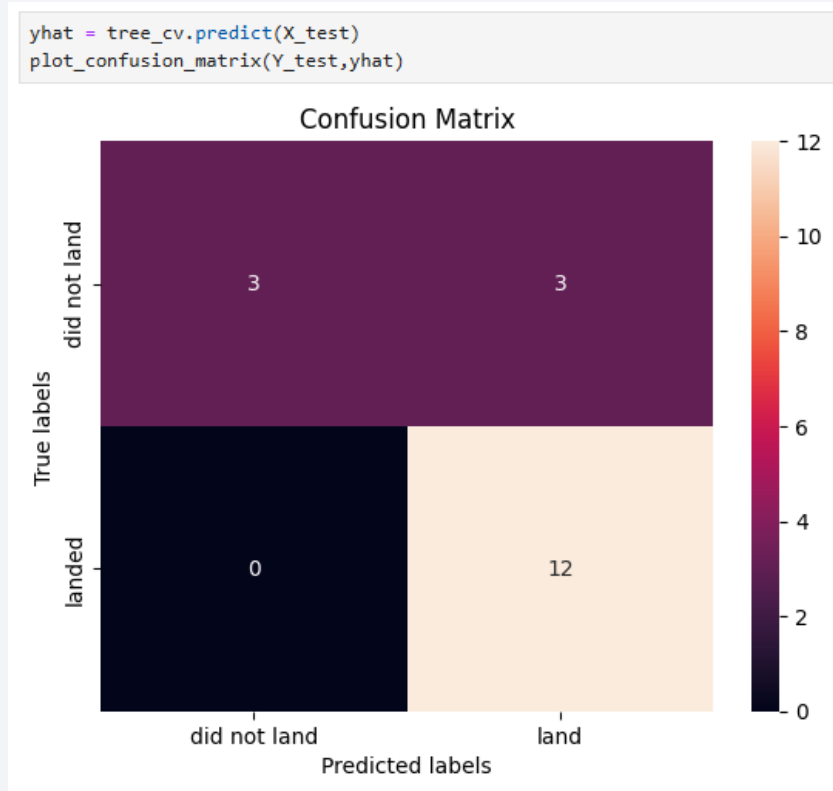
```
[83.33333333333334, 83.33333333333334, 88.88888888888889, 83.33333333333334]  
['LogReg', 'SVM', 'Tree', 'KNN']
```



- Decision Tree outperformed all other models with an accuracy of 88.89%, making it the most reliable classifier in your analysis.
- The other models — Logistic Regression, SVM, and KNN — all tied at 83.33%, showing consistent but slightly lower performance.

Mohammad S. Chowdhury

Confusion Matrix



- The Decision Tree model is highly accurate at predicting successful landings.
- It has zero false negatives, which is crucial when success prediction is mission-critical.
- The false positive rate (3 out of 6 for “did not land”) suggests room for improvement in failure prediction.

Conclusions

- End-to-End Data Science Pipeline Executed - Collected, cleaned, and explored SpaceX launch data using SQL, statistical analysis, and visualizations.
- Business Impact Framed Through Reusability - Focused on predicting first-stage landing success — a key cost driver in SpaceX's competitive pricing model.
- Interactive Dashboards Delivered - Built visual tools to analyze launch sites, payloads, booster versions, and success rates for strategic insights.
- Machine Learning Models Compared - Trained and evaluated multiple classifiers; Decision Tree emerged as the most accurate model (88.89%).
- Actionable Prediction Capability Achieved - Final model enables SpaceY to estimate launch costs and assess reusability potential — bridging data science with aerospace strategy.

Appendix (i)

- Permanent Fix – To clean the database permanently so you don't have to use TRIM every time, you can run an update command:

```
UPDATE SPACEXTBL
```

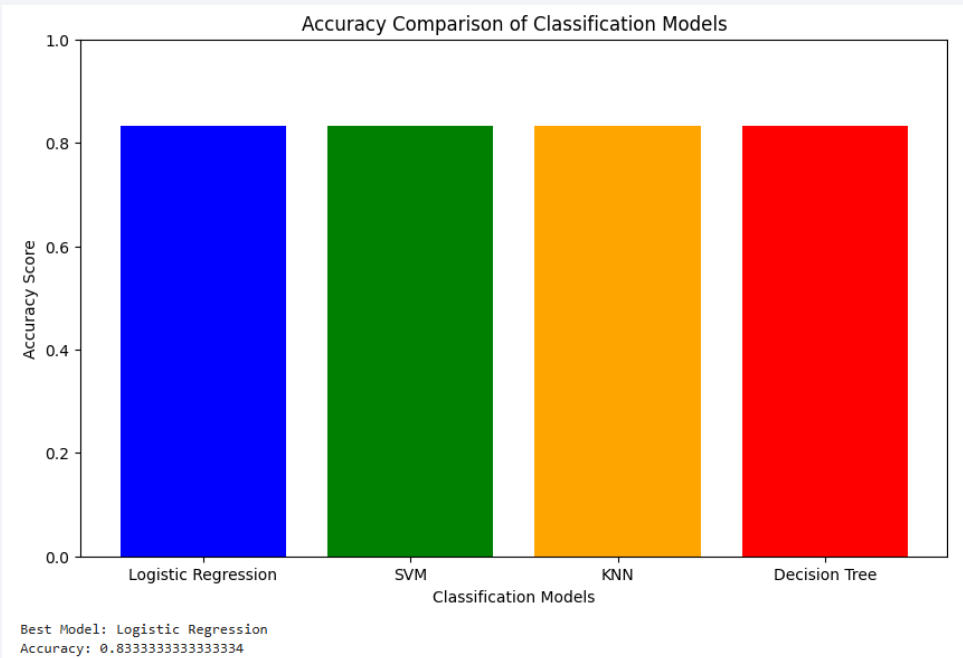
```
SET "Mission_Outcome" = TRIM("Mission_Outcome");
```

Trailing space (e.g., 'Success ' vs 'Success'). SQL treats these as distinct values, which is why they aren't grouping together. By applying TRIM(), both values become identical strings, and the GROUP BY clause will merge them into a single row with a count of 99.

Appendix (ii)

- A Decision Tree might "get lucky" with a specific split previously (0.90), but with a different split, it might be overfitting on noise (0.77).
- The Fix: Ensure you have `random_state=42` (or any integer) in both your split and your model initialization

Appendix (iii)



- Decision Trees have high variance. Small changes in the training data can result in a completely different tree structure.
- Why Logistic Regression won: Logistic Regression is a high bias, low variance model. It is mathematically more "stable".

Appendix (iii)

```
# Pandas is a software library written for the Python programming language for data manipulation and analysis.
import pandas as pd
# NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays
import numpy as np
# Matplotlib is a plotting library for python and pyplot gives us a MatLab like plotting framework. We will use this in our plotter function to plot data.
import matplotlib.pyplot as plt
#Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics
import seaborn as sns
# Preprocessing allows us to standardize our data
from sklearn import preprocessing
# Allows us to split our data into training and testing data
from sklearn.model_selection import train_test_split
# Allows us to test parameters of classification algorithms and find the best one
from sklearn.model_selection import GridSearchCV
# Logistic Regression classification algorithm
from sklearn.linear_model import LogisticRegression
# Support Vector Machine classification algorithm
from sklearn.svm import SVC
# Decision Tree classification algorithm
from sklearn.tree import DecisionTreeClassifier
# K Nearest Neighbors classification algorithm
from sklearn.neighbors import KNeighborsClassifier
np.random.seed(0)
```

- To get the constant result use `np.random.seed(0)` as in the fig.
- Random splitter chooses the best features randomly each time.

Final Closing Summary

This project delivered a complete, end-to-end data science solution that predicts SpaceX first-stage landing success with high accuracy, transforming raw launch data into strategic insights and a deployable decision-making model.

Mohammad S. Chowdhury

Project Executive Wrap-Up

- End-to-end data science pipeline built to analyze and predict Falcon 9 first-stage landing outcomes.
- Integrated data from multiple sources, cleaned and engineered for modeling.
- Dashboards and visual analytics revealed key patterns across sites, payloads, and booster generations.
- Machine learning models trained and compared, with Decision Tree achieving the strongest performance.
- Final predictive model enables cost estimation and strategic planning for a competitive launch provider.

Mohammad S. Chowdhury

Thank you!

Mohammad S. Chowdhury

